# Supplementary Information for FORGE decomposes RNA sequence recovery into memorisation, family transfer and residual prediction error

## Supplementary Note S1: AlphaFold 3 forward-folding control

The forward-folding control evaluates whether the structure-to-sequence information read by FORGE can be inverted by a contemporary folding model. The four solved AI-designed pseudoknot constructs (PDB 10ZT, 10ZU, 11EH and 11AG; design sequences embedded in a circularly permuted group II intron scaffold) were submitted to the AlphaFold 3 server as the full deposited constructs, matching the cryo-EM sample composition (11AG as a two-chain homodimer). Three sequence classes were compared per construct: the true design sequence, the FORGE-derived sequence (per-position maximum of the frozen-model readouts, 36.5–54.2% identical to the true design) and the FORGE-derived sequence after R-Net-guided reactivity refinement (48–59 mutations; reactivity profiles optimised against the cryo-EM secondary structure with FORGE high-confidence positions locked).

Evaluation used the designed segment (96–101 nt) after rigid-body fit to the cryo-EM model. P-RMSD is the mean backbone-phosphate deviation; SS $F_1$ is the $F_1$ of geometric Watson–Crick pairs (donor–acceptor distance $\leq$4.2 Å); contact recovery is the fraction of cryo-EM inter-nucleotide contacts (<7.5 Å) reproduced. One AlphaFold 3 sample was retained per job; the archived batch summaries record a single seed per job.

Only the true 10ZT sequence returned a near-experimental fold (6.6 Å P-RMSD, SS $F_1$ 0.99, 38/38 native pairs recovered). The remaining true sequences failed at 14.1–26.9 Å despite high secondary-structure recovery (SS $F_1$ 0.69–0.88; register shifts within two base pairs recover 32–34 of 32–35 pairs), demonstrating secondary-structure recovery without tertiary-geometry recovery. The FORGE-derived sequences were the worst input class (SS $F_1 \leq 0.13$ on three of four constructs), and R-Net-guided refinement changed the AF3 outcome only marginally (SS $F_1$ 0.00–0.36; the partial exception 11AG improved from 0.13 to 0.36 at 17.3 Å P-RMSD). Sequence recovery read out of geometry therefore does not invert to AF3 consistency, with or without reactivity-guided sequence optimisation.

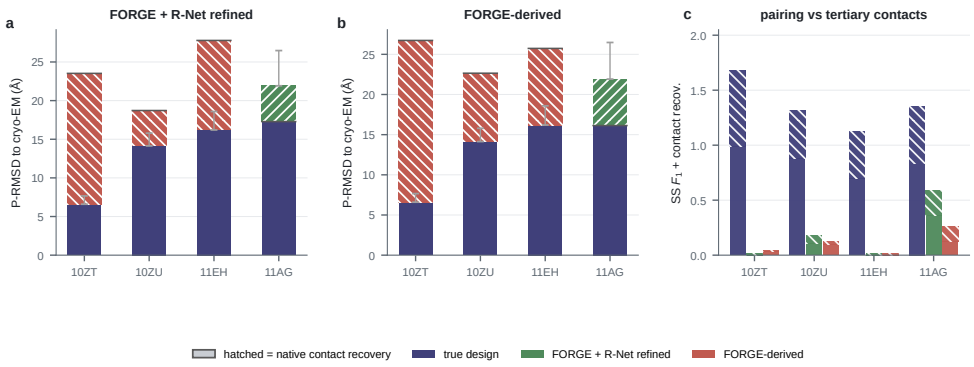



**Fig. S1 Forward-folding evaluation of the AI-designed pseudoknot constructs (AlphaFold 3). a,** Delta P-RMSD of the FORGE-derived sequence after R-Net-guided refinement relative to the true design sequence (hatched segments: green, improvement; red, degradation; grey error bars, range across archived server outputs for the true sequence; one sample per job). **b,** Same for the FORGE-derived sequence. **c,** Watson–Crick pair recovery (SS $F_1$, solid) and native tertiary contact recovery (hatched) for the three sequence classes.

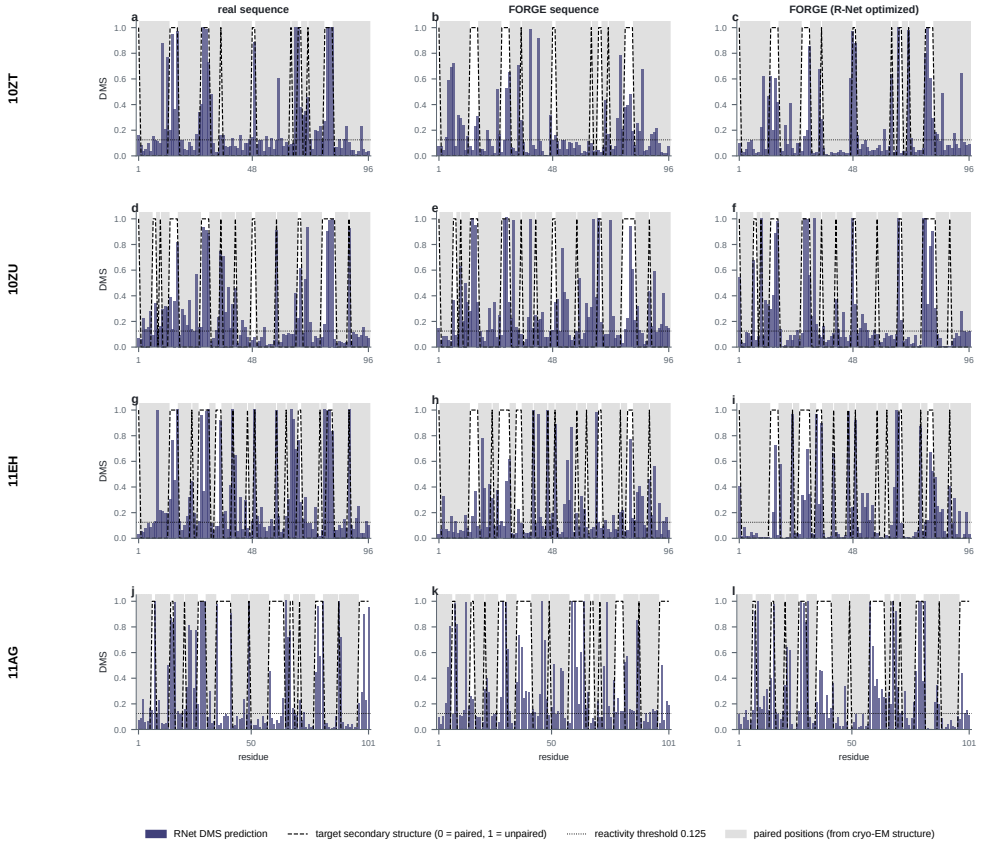



**Fig. S2 R-Net reactivity predictions against the target secondary structure.** Rows, PDB constructs; columns, sequence classes (true design, FORGE-derived, FORGE-derived after R-Net refinement). Grey bands mark positions paired in the cryo-EM structure; the dashed line is the binary target profile (0, paired; 1, unpaired); the dotted line is the reactivity threshold 0.125.

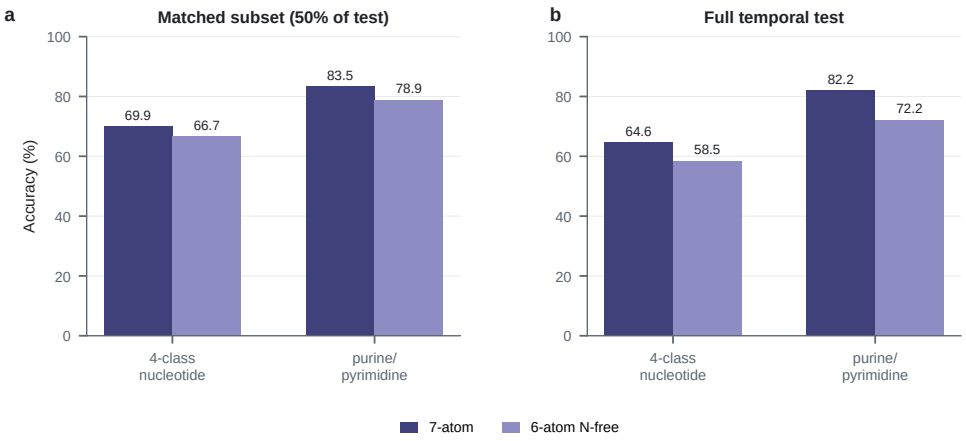



**Fig. S3 N-free retraining controls at both evaluation scales. a,** Matched-subset retraining (200,000 training positions, 72,787 test positions): the seven-atom model reached 69.9% four-class recovery and 83.5% purine/pyrimidine accuracy, and the six-atom N-free model 66.7% and 78.9%. **b,** Full temporal comparison: the seven-atom model on 4,135 chains (291,331 positions) reached 64.6% weighted recovery and 82.2% purine/pyrimidine accuracy; the N-free model on 4,167 chains (298,662 positions) reached 58.5% and 72.2%. Both scales show that most nucleotide-discriminating signal persists without the glycosidic nitrogen coordinate.




**Fig. S4 Confidence and chemical-class behaviour on the solved AI-designed pseudoknots. a,** Mean per-position FORGE confidence per chain (full post-2025 reference, 0.48). **b,** Fraction of positions with confidence >0.7, annotated with the absolute number of high-confidence positions. **c,** Purine/pyrimidine accuracy for the seven-atom and six-atom N-free models (dashed line, random 50%).

# Supplementary Table S1: Evaluation subsets

| Analysis | Chains | Positions | Selection Criterion | Metric Type | Key Value |
|---|---|---|---|---|---|
| Full post-2025 temporal | 4,135 | 291,331 | All post-2025 RNA chains | Weighted (micro) | 64.6% [63.7, 65.6] |
| Full post-2025 temporal (per-chain) | 4,135 | 291,331 | All post-2025 RNA chains | Per-chain mean (macro) | 63.4% |
| Exact-sequence repeat | 1,281 | 86,729 | Exact sequence in pre-2025 training | Weighted (micro) | 87.1% [86.1, 88.2] |
| Sequence-novel | 2,854 | 204,602 | Exact sequence absent from training | Weighted (micro) | 55.1% [54.3, 55.9] |
| Calibration (full) | 4,135 | 291,331 | All post-2025 RNA chains | ECE, 10 bins | 0.023 [0.020, 0.026] |
| In-domain reference | 591 | 53,288 | Sequence/PDB-disjoint held-out components | Weighted (micro) | 70.0% |
| Temporal on same chains | 591 | 53,288 | Same held-out chains | Weighted (micro) | 54.0% |
| Family-disjoint in-domain | 410 | 32,984 | Sequence/PDB/family-disjoint held-out components | Weighted (micro) | 50.4% |
| Temporal on family-disjoint chains | 410 | 32,984 | Same family-disjoint chains | Weighted (micro) | 47.5% |
| Cluster-disjoint temporal | 3,749 | 267,477 | No shared pre-2025 CD-HIT cluster | Weighted (micro) | 64.3% [63.3, 65.3] |
| Novel and cluster-disjoint | 2,587 | 190,127 | Exact-novel and cluster-disjoint | Weighted (micro) | 55.2% [54.4, 56.0] |
| Homology <80% identity | 2,078 | 146,058 | Max identity to training <80% | Weighted (micro) | 48.9% |
| N-removed matched | 4,135 | 291,331 | Same positions as 7-atom | Weighted (micro) | 58.5% [57.4, 59.6] |
| Stacking | 4,135 | 290,878 | Valid base-plane positions | Acc./AUC | 81.8% / 0.879 |
| SS class | 2,138 | 163,395 | Exact DSSR-length chains | Acc./F1 | 78.5% / 0.542 |
| Solvent exposure | 4,135 | 291,331 | Post-2025 chains with full-atom SASA labels | Pearson / $R^2$ | 0.948 / 0.899 |
| N-removed archived | 4,167 | 298,662 | Archived coverage | Weighted (micro) | 58.5% |
| Native-vs-decoy | 50 | — | 20–200 nt, baseline designs available | Chain-level AUROC | 0.41–1.00 by tier |
| Stress test | 5 chains | 490 | AI-designed pseudoknots, cryo-EM | Per-chain | 36.5–54.2% |
| $k$-NN feature-space | 40 query | 1,134 | 10–200 nt, ref pool 80 chains | Position (micro) | $k$=10: 53.6% |
| OpenKnot complete-case | 1,656 | — | MFE F1 available | Design-level $\rho$ | $\rho_{\text{partial}}$=+0.549 |

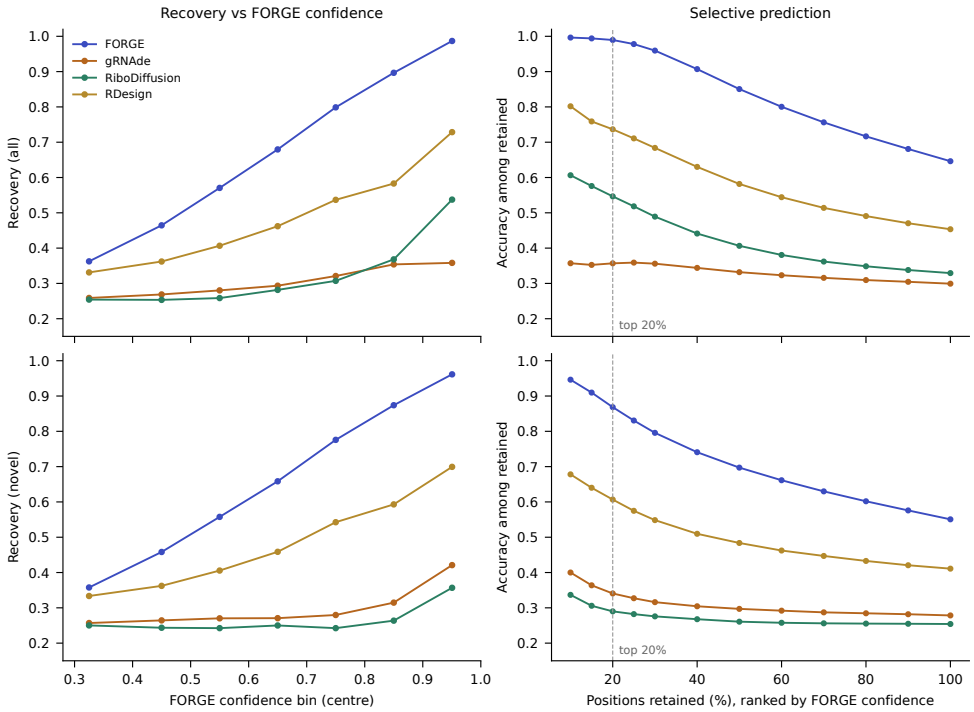



**Fig. S5 Inverse-folding recovery tracks model-based predictability but falls far below it.** Positions are ranked by FORGE confidence. **a,** Recovery across confidence bins for FORGE and three public inverse-folding models on all post-2025 positions (top) and on sequence-novel positions only (bottom). **b,** Selective prediction: accuracy on the retained top-$k$ fraction of positions. On sequence-novel positions, the rank-defined top 20% of sequence-novel positions (40,920 positions) are recovered at 86.9% by FORGE but only 60.7% (RDesign), 34.1% (gRNAde) and 29.0% (RiboDiffusion); this rank-defined subset is distinct from the confidence $\geq$0.9 bin used in Supplementary Table S6. Values are computed on the matched position mask; public models were not retrained on the FORGE split.

Calibration used 10 equal-width confidence bins from 0.25 to 1.0 on all 291,331 positions (ECE 0.023, Brier 0.304, negative log-likelihood 0.826; 500 chain-bootstrap resamples). Per-chain abstention is right-censored at 90% removal and unstable for short chains; those estimates are descriptive only. A protein-proximal context analysis was explored but is not reported as a result because chain-aware residue joins and negative sampling were not independently validated. Synthetic tests confirmed that the Shrake–Rupley implementation reproduces the analytic isolated-atom area and gives identical labels after a [100, 200, 300] Å translation. A label-shuffle control trained on permuted SASA labels gave r = $-0.011$ ($R^2 < 0$), confirming that the solvent-exposure signal is not an artifact of the regression pipeline. The SASA readout uses the same cluster-permissive temporal test as the nucleotide readout and is not a novelty-stratified estimate; residues with non-standard atom mappings are a small minority and may carry noisier labels. Figure 2a uses matched-position values (3,989/3,966/3,035 matched chains); the all-position published values (30.1/33.5/46.4) are retained only for provenance and are not a controlled benchmark.

# Supplementary Table S2: Stress-test chain homology verification

| Chain | Length | In training? | Best seq. identity | Best match | PDB deposit | Download date |
|---|---|---|---|---|---|---|
| 10ZT (gRNAde) | 96 | No | 36.0% | 5O9Z_X_2 (100 nt) | 2026-05-13 | 2026-05-24 |
| 10ZU (MPNN-fixbb) | 96 | No | 38.5% | 3ADC_H_C (88 nt) | 2026-05-13 | 2026-05-24 |
| 11EH (Struct2Seq) | 96 | No | 34.4% | 3A2K_H_C (77 nt) | 2026-05-14 | 2026-05-24 |
| 11AG A (MPNN-RFdiff) | 101 | No | 35.6% | 6CK4_H_D (101 nt) | 2026-05-14 | 2026-05-24 |
| 11AG C (MPNN-RFdiff) | 101 | No | 35.6% | 6CK4_H_D (101 nt) | 2026-05-14 | 2026-05-24 |

# Supplementary Table S3: Multi-task label provenance and leakage control

| Task | Label source | Forbidden inputs removed | Split | Baseline |
|---|---|---|---|---|
| Nucleotide identity | PDB sequence | None; 7-atom coordinates only | Pre-2025 / post-2025 temporal | 25% random |
| Base-pair state | DSSR dot-bracket | No DSSR, dot-bracket or pair labels used as features; all 935 descriptors computed from coordinates alone | Pre-2025 / post-2025 temporal | 63.1% majority class |
| Reactivity target | RibonanzaNet inference | Sequence not used as input; backbone coordinates only | Pre-2025 / post-2025 temporal | Mean predictor |
| Solvent exposure | Full-atom Shrake–Rupley rSASA | Full-atom coordinates used only for labels; seven-atom input | Pre-2025 train / post-2025 test | Training median 0.395 |
| Stacking | Full-atom base-plane label | Sign-aligned normals; invalid planes masked | Pre-2025 train / post-2025 test | 6 Å, 35°, 3.5 Å |
| SS class | DSSR dot-bracket | Exact length agreement; 3 classes | Pre-2025 train / post-2025 test | class-balanced weights |

# Supplementary Table S4: Retrained atom-representation controls

| Model | Atom channels used | Columns removed | Weighted recovery | Δ vs 7-atom (pp) |
|---|---|---|---|---|
| 7-atom control | P, O5′, C5′, C4′, C3′, O3′, N1/N9 | 0 | 61.94% | — |
| 6-atom, N-neutralised | six; N-dependent features removed | 164 | 58.50% | −3.43 [−3.65, −3.23] |
| 6-atom, N at origin | six; N channel set to origin | 0 | 57.25% | −4.69 [−4.97, −4.42] |
| 6-atom, N = C4′ | six; dummy anchor | 0 | 57.42% | −4.52 [−4.81, −4.24] |
| 5-atom | O5′/O3′ features removed | 257 | 61.63% | −0.31 [−0.47, −0.16] |
| 3-atom | O5′/C5′/C3′/O3′ features removed | 435 | 61.04% | −0.90 [−1.08, −0.71] |

# Supplementary Table S5: Sequence-homology audit

| Maximum identity to training | Chains | Positions | Definition | Weighted recovery |
|---|---|---|---|---|
| ≥95% | 1,580 | 112,697 | Best global identity ≥95% | 85.0% |
| 90–95% | 219 | 13,066 | Best identity 90–95% | 67.4% |
| 80–90% | 258 | 19,510 | Best identity 80–90% | 62.8% |
| <80% | 2,078 | 146,058 | No training chain at ≥80% identity | 48.9% |

Identity is the maximum global Needleman–Wunsch identity to any pre-2025 training chain (edlib, 12-mer prefilter, 2,500-candidate cap; exact matches assigned 1.0).

# Supplementary Table S6: Multi-learner convergence

| FORGE confidence | Positions | XGBoost | Logistic | MLP | $k$NN |
|---|---|---|---|---|---|
| <0.5 | 118,156 | 41.7% | 27.4% | 31.8% | 25.8% |
| 0.5–0.7 | 70,468 | 61.3% | 33.1% | 42.1% | 32.9% |
| 0.7–0.8 | 19,941 | 79.9% | 41.6% | 60.5% | 51.8% |
| 0.8–0.9 | 19,268 | 89.7% | 48.5% | 75.1% | 66.6% |
| ≥0.9 | 63,498 | 98.7% | 64.3% | 94.5% | 91.5% |
| Overall | 291,331 | 64.6% | 39.2% | 52.8% | 46.3% |

Logistic regression (SGD, standardized), MLP (512–256, early stopping) and $k$NN ($k = 10$) were trained on the same 200,000-position subsample; $k$NN was evaluated on a 20,000-position query subsample.

## Supplementary Table S7: Frame-invariance audit

| Transform | Prediction agreement | Recovery | Note |
|---|---|---|---|
| Original | — | 59.3% | 120 chains, 7,790 positions |
| Translation | 97.9% | 59.5% | translation-robust |
| Rotation | 59.0% | 47.9% | not rotation-invariant |
| Rigid (rotation + translation) | 58.2% | 47.8% | not rigid-invariant |
| Rotation-masked retrained model | 53.3% | 62.1% | 112 rotation-dependent features removed; agreement under rotation remains low |
| Canonicalised-frame model | 99.6% | 62.3% | centred and principal-axis aligned before featurization; repeats 84.7%, exact-novel 52.8%, <80% identity 47.0%, ECE 0.012 |

The frame audit shows that the current pipeline is translation-robust but not rotation-invariant. A model retrained without the 112 rotation-dependent features retains 62.1% unrotated recovery but only 53.3% agreement under rotation, so it does not restore invariance. The canonicalised-frame model (62.3% recovery, 99.6% agreement) reproduces the decomposition (84.7% repeats, 52.8% exact-novel, 47.0% below 80% identity; ECE 0.012) with a paired cost of 2.35 pp, so the main conclusions are frame-safe at a small cost; the headline estimates themselves apply to the locked coordinate convention.

## Supplementary Table S8: within-family method and resolution strata

| Family | Method | Resolution (Å) | Chains | Positions | Recovery |
|---|---|---|---|---|---|
| tRNA | cryo-EM | <=2.5 | 101 | 7,525 | 79.7% |
| tRNA | cryo-EM | 2.5-3.0 | 178 | 13,327 | 71.7% |
| tRNA | cryo-EM | 3.0-3.5 | 95 | 7,031 | 70.1% |
| tRNA | cryo-EM | >3.5 | 84 | 6,372 | 83.0% |
| tRNA | X-ray | <=2.5 | 37 | 2,717 | 86.5% |
| tRNA | X-ray | 2.5-3.0 | 44 | 3,253 | 77.2% |
| tRNA | X-ray | 3.0-3.5 | 1 | 88 | 73.9% |
| 5S_rRNA | cryo-EM | <=2.5 | 85 | 10,106 | 86.2% |
| 5S_rRNA | cryo-EM | 2.5-3.0 | 126 | 14,979 | 84.9% |
| 5S_rRNA | cryo-EM | 3.0-3.5 | 71 | 8,440 | 87.7% |
| 5S_rRNA | cryo-EM | >3.5 | 51 | 6,125 | 92.8% |
| 5S_rRNA | X-ray | <=2.5 | 10 | 1,200 | 100.0% |
| 5S_rRNA | X-ray | 2.5-3.0 | 22 | 2,644 | 100.0% |
| 5_8S_rRNA | cryo-EM | <=2.5 | 33 | 5,148 | 90.2% |
| 5_8S_rRNA | cryo-EM | 2.5-3.0 | 70 | 10,775 | 95.3% |
| 5_8S_rRNA | cryo-EM | 3.0-3.5 | 27 | 4,165 | 95.7% |
| 5_8S_rRNA | cryo-EM | >3.5 | 12 | 1,865 | 97.8% |
| Purine | cryo-EM | 2.5-3.0 | 5 | 703 | 43.7% |
| Purine | X-ray | <=2.5 | 5 | 346 | 40.5% |
| Purine | X-ray | 2.5-3.0 | 9 | 637 | 37.0% |
| Purine | X-ray | 3.0-3.5 | 3 | 211 | 34.1% |
| RNaseP_bact_b | cryo-EM | 2.5-3.0 | 42 | 17,514 | 50.7% |
| RNaseP_bact_b | cryo-EM | 3.0-3.5 | 34 | 14,178 | 51.3% |
| RNaseP_bact_b | cryo-EM | >3.5 | 2 | 834 | 48.6% |
| AdoCbl-II | X-ray | <=2.5 | 24 | 1,800 | 61.8% |
| AdoCbl-II | X-ray | 2.5-3.0 | 5 | 376 | 63.3% |

Within-family strata are descriptive; families with fewer than five chains in a bin are omitted. The non-monotonic pattern (for example, cryo-EM tRNA chains above 3.5 Å recover better than 3.0–3.5 Å chains) is consistent with family and length confounding and is why the adjusted model, not the raw bins, is used for inference.

# Supplementary Figure S6: design-objective separability and stress tests

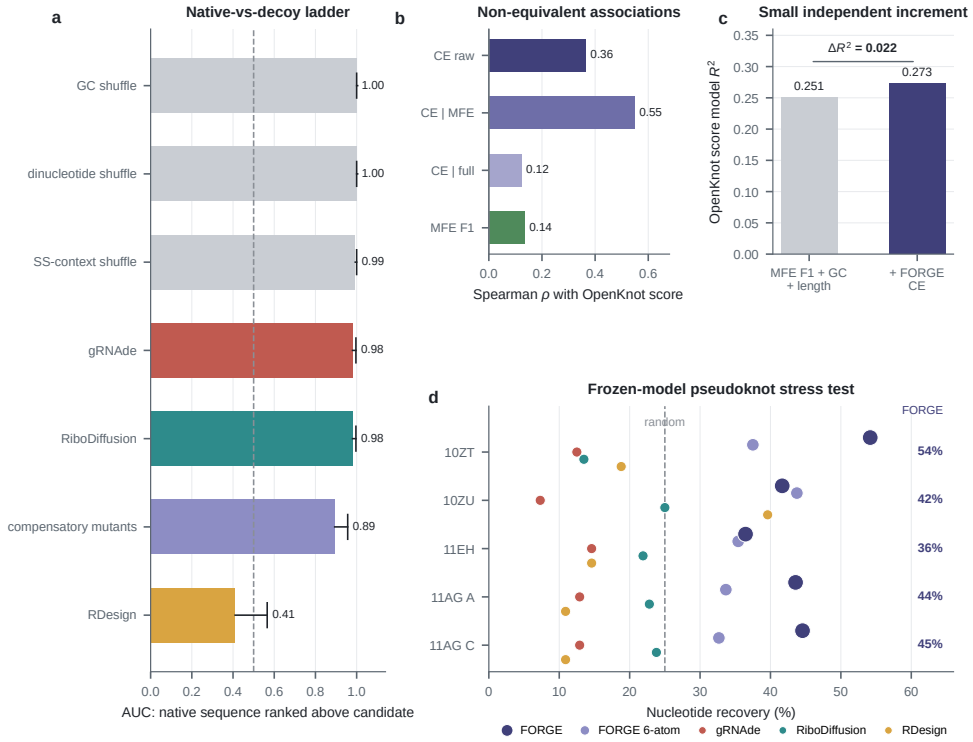



**Fig. S6 Design-objective separability and frozen-model stress tests. a,** Native-versus-decoy AUC ladder across composition-preserving shuffles, learned inverse-folding outputs, compensatory pair swaps and RDesign outputs (error bars, Wilson 95% intervals from stored tier aggregates; dashed line, no native preference). **b,** Raw and partial Spearman associations between FORGE cross-entropy, RNAfold MFE F1 and OpenKnot experimental score in the complete-case design subset ($n = 1,656$). **c,** Linear-model fit to OpenKnot score before and after adding FORGE cross-entropy to MFE F1, GC content and length ($\Delta R^2 = 0.022$, $p = 4.3 \times 10^{-12}$). **d,** Per-chain nucleotide recovery (dot plot) for FORGE, the six-atom N-free retrained model and public pretrained baselines on five solved AI-designed pseudoknot chains; the dashed line marks random recovery.

# Supplementary Figure S7: representation controls and feature-space ambiguity

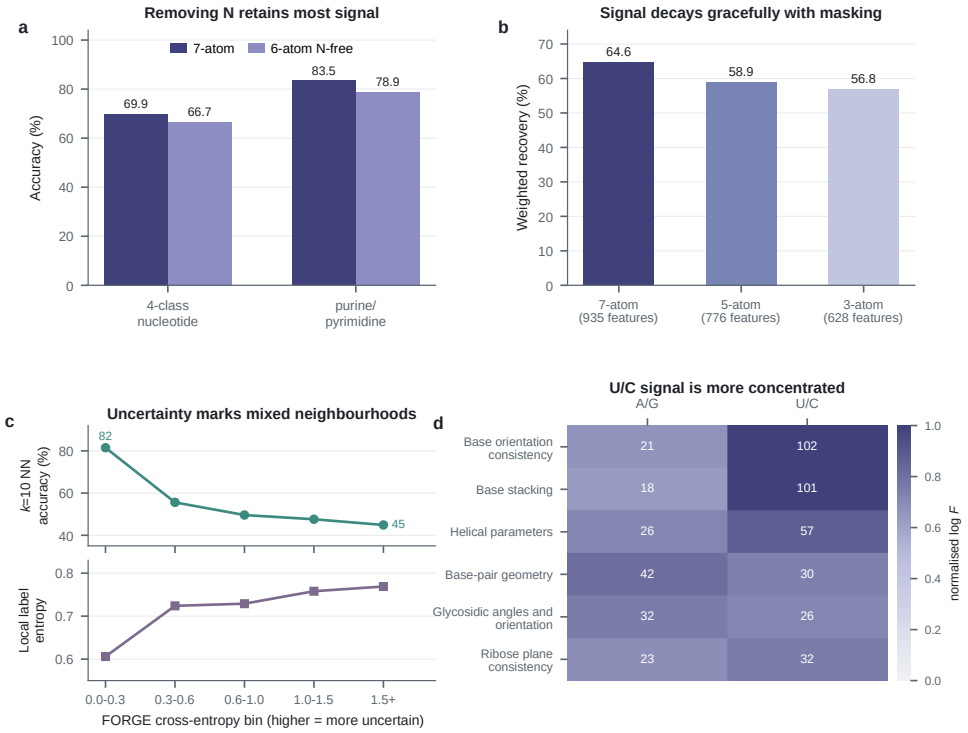



**Fig. S7 Representation controls and feature-space ambiguity. a,** Matched-subset retraining after N-channel neutralisation (4-class recovery and purine/pyrimidine accuracy; full comparison 64.6% versus 58.5%). **b,** Inference-time atom-feature masking on the full post-2025 set (7-, 5- and 3-atom masks with the number of retained features). **c,** Leave-one-chain-out $k = 10$ nearest-neighbour accuracy (top) and local label entropy (bottom) across FORGE cross-entropy bins. **d,** Feature-ontology categories with the strongest discrimination for the A/G and U/C ambiguous pairs (colour, normalised log-transformed ANOVA $F$-statistic; numbers, raw $F$).

# Supplementary Methods: retrospective strata and feature attribution

### *Structure metadata.*

Resolution and experimental method were retrieved for all 1,517 distinct post-2025 PDB entries from the RCSB entry API on 2026-10-30 (`pdb_metadata.json`). Resolution was taken from the first value of `resolution_combined`; entries without

a numeric resolution were kept in an explicit missing/NMR category for descriptive strata and were excluded from the adjusted OLS model. X-ray and cryo-EM resolutions were binned separately and never pooled without method adjustment.

***Adjusted model.***

Chain-level recovery was modelled as `recovery ~ log(length) + method + resolution + family + release year + novelty` by OLS with PDB-clustered standard errors, using the 4,084 chains with numeric resolution (51 missing-resolution chains excluded). The resolution coefficient was +1.1 pp per Å (95% CI −0.2 to +2.5; $p = 0.10$) and the X-ray-versus-cryo-EM term was +1.8 pp ($p = 0.26$); exact-novel, cluster-disjoint chains recovered 17.7 pp less ($p < 10^{-24}$) and 2026 entries 6.3 pp more than 2025 entries ($p < 10^{-5}$). The outcome is bounded in [0,1]; OLS coefficients are associational and may be attenuated. Family terms used the six groups with at least 50 chains; smaller families were pooled as "other". The descriptive family panel uses families with at least 20 chains. These estimates are association-only and do not identify causal quality effects.

***Grouped permutation importance.***

The 935 descriptors were mapped to the six feature-ontology categories (914 mapped; 21 unclassified). For each task and category, all columns in the category were permuted with a common row permutation, preserving within-category correlations, on a fixed 20,000-position post-2025 sample; the metric drop was averaged over five fixed-seed permutations. Nucleotide identity used accuracy, base-pair state AUC and solvent accessibility Pearson $r$ (not $R^2$). Point estimates are mean per-chain drops; 95% intervals are chain bootstrap (2,000 resamples); five fixed permutations; absolute drops and the five-permutation range are exploratory and conditional on the frozen model, and are in the source CSV. The frozen models were not retrained. Groups differ substantially in size (14–693 features); drops are group-level and are not normalised per feature.

***SASA ablation controls.***

The solvent-accessibility regressor was retrained on the same 200,000-position pre-2025 sample after setting to zero the four explicit `sasa_*` features, the 24 global/centre descriptors, the nine local-density features, or all three groups (35 features). Pearson $r$ fell from 0.948 (all features) to 0.948 (no explicit SASA), 0.942 (no SASA + global) and 0.942 (no SASA + global + density), showing that the signal is distributed across local geometry rather than carried by a single proxy. Zeroing is an ablation, not physical feature removal; it supports packing-associated coordinate predictability, not independent chemistry.

***Case-study selection.***

Selection rules were documented before chain-level outcomes were inspected; A is a confidence-selected positive control and B is family-targeted: (A) lexicographically first X-ray chain at $\leq 2.0$ Å or cryo-EM chain at $\leq 3.0$ Å, length 50–300 nt, at least 30% of positions at confidence $\geq 0.9$, family in 5S rRNA, 5.8S rRNA or tRNA; (B)

lexicographically first Purine or K10_TLS chain, length 50–300 nt; (C) lexicographically first post-2025 chain with at least 10 DSSR pseudoknot positions and length 50–300 nt; (D) the median-length chain among the first 100 lexicographic exact-novel, cluster-disjoint chains. Full eligible lists (425/22/146/100 chains for A–D) and selected IDs are archived in `case_selection_full.json`. Rule A is a confidence-selected positive control; Rule B is family-targeted (Purine/K10_TLS, known at the family level to be low-recovery); Rules C/D are outcome-blind. 10ZT/10ZU/11EH/11AG are not in the locked post-2025 feature set, so Rule C uses a DSSR pseudoknot chain. These provenance details are descriptive and do not support statistical inference.

## Supplementary Table S9: complete multiplicity table

| Analysis family | Test | Estimate | $p$ | BH $q$ | Status |
|---|---|---|---|---|---|
| Family | 5.8S_rRNA | +33.9 pp | $8.5 \times 10^{-131}$ | $3.4 \times 10^{-130}$ | adjusted |
| Family | 5S_rRNA | +24.9 pp | $9.9 \times 10^{-76}$ | $2.0 \times 10^{-75}$ | adjusted |
| Family | RNaseP | +1.0 pp | 0.496 | 0.496 | adjusted |
| Family | tRNA | +18.6 pp | $9.2 \times 10^{-63}$ | $1.2 \times 10^{-62}$ | adjusted |
| Length | <50 | +0.4 pp | 0.759 | 0.759 | adjusted |
| Length | 50-100 | +0.5 pp | 0.732 | 0.759 | adjusted |
| Length | 201-500 | -1.8 pp | 0.430 | 0.759 | adjusted |
| Resolution | <=2.0 | -6.0 pp | 0.059 | 0.149 | adjusted |
| Resolution | 2.0-2.5 | +3.4 pp | 0.147 | 0.244 | adjusted |
| Resolution | 3.0-3.5 | -0.2 pp | 0.911 | 0.911 | adjusted |
| Resolution | 3.5-4.0 | +1.7 pp | 0.506 | 0.633 | adjusted |
| Resolution | >4.0 | +5.4 pp | $7.1 \times 10^{-4}$ | 0.004 | adjusted |
| Feature group | Intra-nt backbone (nucleotide) | 20.4% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Global/centre (nucleotide) | 2.2% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Unclassified (nucleotide) | 7.9% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Helical/torsion (nucleotide) | 2.0% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Local neighborhood (nucleotide) | 1.4% | 0.002 | 0.002 | exploratory |
| Feature group | Glycosidic (nucleotide) | 2.0% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Inter-nt edge (nucleotide) | 30.6% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Intra-nt backbone (basepair) | 0.1% | 0.557 | 0.561 | exploratory |
| Feature group | Global/centre (basepair) | 2.4% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Unclassified (basepair) | 0.1% | 0.145 | 0.160 | exploratory |
| Feature group | Helical/torsion (basepair) | -0.1% | 0.561 | 0.561 | exploratory |
| Feature group | Local neighborhood (basepair) | 0.3% | 0.144 | 0.160 | exploratory |
| Feature group | Glycosidic (basepair) | 0.4% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Inter-nt edge (basepair) | 23.9% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Intra-nt backbone (sasa) | 3.9% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Global/centre (sasa) | 31.5% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Unclassified (sasa) | 0.4% | 0.002 | 0.002 | exploratory |
| Feature group | Helical/torsion (sasa) | 2.1% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Local neighborhood (sasa) | 10.0% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Glycosidic (sasa) | 2.4% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Feature group | Inter-nt edge (sasa) | 12.2% | $1.0 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | exploratory |
| Covariate | novelty | – | $2.6 \times 10^{-25}$ | $5.3 \times 10^{-25}$ | adjusted |
| Covariate | year_2026_vs_2025 | – | $9.7 \times 10^{-6}$ | $9.7 \times 10^{-6}$ | adjusted |
| SASA threshold | threshold 0.3 | AUC 0.971 | – | – | descriptive |
| SASA threshold | threshold 0.35 | AUC 0.946 | – | – | descriptive |
| SASA threshold | threshold 0.4 | AUC 0.940 | – | – | descriptive |
| SASA threshold | threshold 0.45 | AUC 0.979 | – | – | descriptive |
| SASA threshold | threshold 0.5 | AUC 0.980 | – | – | descriptive |
| SASA threshold | threshold 0.55 | AUC 0.990 | – | – | descriptive |
| SASA threshold | threshold 0.6 | AUC 0.993 | – | – | descriptive |

$p$-values for family, resolution and length contrasts come from PDB-clustered OLS with the other strata as covariates; feature-group $p$-values are two-sided chain-bootstrap values; covariates are the novelty and 2026-vs-2025 terms. BH correction

is applied within each declared family. SASA thresholds are descriptive and carry no $p$-value. Feature-group $p$-values are bootstrap sign probabilities after five permutations (floor $10^{-6}$) and are exploratory, not permutation-null tests. The five sensitivity terms are corrected separately in Table S10b.

## Supplementary Table S10: model sensitivity and importance controls

| Resolution sensitivity | Estimate | $p$ |
|---|---|---|
| Linear resolution (pp/Å) | +1.13 | 0.099 |
| Logit OLS (logit/Å) | +0.143 | 0.119 |
| Fractional logit (logit/Å) | +0.054 | 0.107 |
| X-ray only (pp/Å) | +4.54 | 0.016 |
| Spline joint test | – | $2.8 \times 10^{-4}$ |
| Quadratic term | – | 0.830 |
| Method x resolution | – | 0.119 |
| Family x resolution | – | 0.017 |

Missing resolution: 51 chains (mean recovery 45.5%); included $n = 4{,}084$, of which 1,411 carry an Rfam label. Maximum Cook's distance 0.0053; maximum VIF 4.83; 1,472 PDB clusters. Grouped SHAP uses 5,000 positions per task; group sizes 14–693 and shares sum to 100%.

| Leave-one-group-out (pp change) | Nucleotide | Base-pair | SASA |
|---|---|---|---|
| Intra-nt backbone | +2.20 | +0.11 | -0.01 |
| Global/centre | +0.39 | +0.33 | +0.57 |
| Unclassified | -0.24 | +0.02 | -0.04 |
| Helical/torsion | -0.43 | +0.10 | +0.04 |
| Local neighborhood | -0.28 | +0.04 | +0.06 |
| Glycosidic | +0.02 | +0.08 | +0.09 |
| Inter-nt edge | +1.65 | +0.89 | +0.87 |

Label-shuffle controls (training labels permuted): nucleotide accuracy 26.2% (chance 25%), base-pair AUC 0.502 (chance 0.5), SASA Pearson $r = -0.011$ (chance 0). These are pipeline sanity checks; they do not test train/test homology or coordinate shortcuts, which are covered by the temporal/cluster audits. A single shuffle seed is a chance-level control, not a calibrated null.

| Grouped SHAP share (%) | Nucleotide | Base-pair | SASA |
|---|---|---|---|
| Intra-nt backbone | 18.8 | 6.6 | 9.3 |
| Global/centre | 5.3 | 12.4 | 27.9 |
| Unclassified | 6.0 | 4.4 | 1.7 |
| Helical/torsion | 5.9 | 6.0 | 3.7 |
| Local neighborhood | 7.4 | 7.8 | 15.5 |
| Glycosidic | 3.7 | 4.1 | 3.3 |
| Inter-nt edge | 52.8 | 58.8 | 38.7 |

## Supplementary Table S10b: exploratory sensitivity terms

| Sensitivity term | Estimate | $p$ | BH $q$ |
|---|---|---|---|
| Spline joint test | – | $2.8 \times 10^{-4}$ | 0.0014 |
| X-ray-only resolution | +4.54 pp/Å | 0.016 | 0.028 |
| Quadratic resolution | – | 0.830 | 0.830 |
| Method × resolution | – | 0.119 | 0.149 |
| Family × resolution | – | 0.017 | 0.028 |

BH correction is across these five exploratory sensitivity terms and does not replace the primary strata contrasts in Table S9. X-ray-only is the resolution slope within X-ray chains; the adjusted X-ray-versus-cryo-EM method term (+1.8 pp) is a distinct estimand from the main model.

## Supplementary Table S11: SASA threshold and novelty sensitivity

| Exposure threshold | Exposed fraction | AUC | Average precision |
|---|---|---|---|
| 0.3 | 0.834 | 0.971 | 0.993 |
| 0.35 | 0.690 | 0.946 | 0.973 |
| 0.4 | 0.424 | 0.940 | 0.933 |
| 0.45 | 0.270 | 0.979 | 0.954 |
| 0.5 | 0.173 | 0.980 | 0.925 |
| 0.55 | 0.110 | 0.990 | 0.947 |
| 0.6 | 0.077 | 0.993 | 0.938 |

| Population | Chains | Positions | Length-weighted mean per-chain $r$ | 95% CI |
|---|---|---|---|---|
| All | 4,135 | 291,331 | 0.922 | [0.920, 0.924] |
| Exact repeat | 1,281 | 86,729 | 0.954 | [0.951, 0.957] |
| Exact novel | 2,854 | 204,602 | 0.909 | [0.906, 0.911] |
| <80% identity | 2,078 | 146,058 | 0.903 | [0.901, 0.905] |
| Novel + cluster-disjoint | 2,587 | 190,127 | 0.910 | [0.908, 0.912] |

AUC remains high (0.940–0.993) but is threshold-dependent; the novelty values are length-weighted means of per-chain $r$, not the pooled $r = 0.948$. SASA remains a coordinate-derived packing proxy on the cluster-permissive temporal population.

## Supplementary Figure S12: complementary descriptor and stratum panels

## Supplementary Table S13: within-puzzle OpenKnot analysis

| Statistic | Value |
|---|---|
| Designs / puzzles | 8,125 / 21 |
| Pooled Spearman | 0.289 |
| Within-puzzle normalized rank correlation | 0.078 |
| Held-out-target Spearman, mean (range) | 0.088 (−0.363 to +0.513) |
| Leave-one-puzzle-out pooled range | 0.171–0.330 |
| Per-puzzle Spearman range | −0.363 to +0.513 |
| Held-out MFE+GC+length baseline $\rho$ | 0.316 |
| Held-out + FORGE CE $\rho$ | 0.332 |
| Held-out $\Delta\rho$, positive puzzles | +0.016; 13/21 |

The held-out baseline merge uses the stable key (puzzle, sequence, method, round), with duplicate keys aggregated by mean; 8,125 scored and 8,125 MFE rows collapse to 7,837 unique keys (no many-to-many amplification). The pooled association is not evidence of target-generalizing design ranking: normalized within-puzzle correlation is weak, held-out-target prediction is mixed, and leave-one-puzzle-out is a deletion sensitivity, not a held-out validation.

## Supplementary Figure S14: ligand context and NMR ensemble flexibility

***Ligand-context analysis.***

The 200 post-2025 chains in `binding_annotations_v2.json` provide per-position ligand/ion/water distances; we mapped annotations to residue positions, added FORGE predictions and compared mean confidence and accuracy by binding class. Classes with fewer than five positions are not interpreted. The analysis is descriptive and does not establish causal ligand effects. All 200 chain lengths match their FASTA; an in-range join retained 17,678 of 17,991 annotations (313 excluded as out-of-range or non-A/C/G/U). This checks position bounds and A/C/G/U identity against the

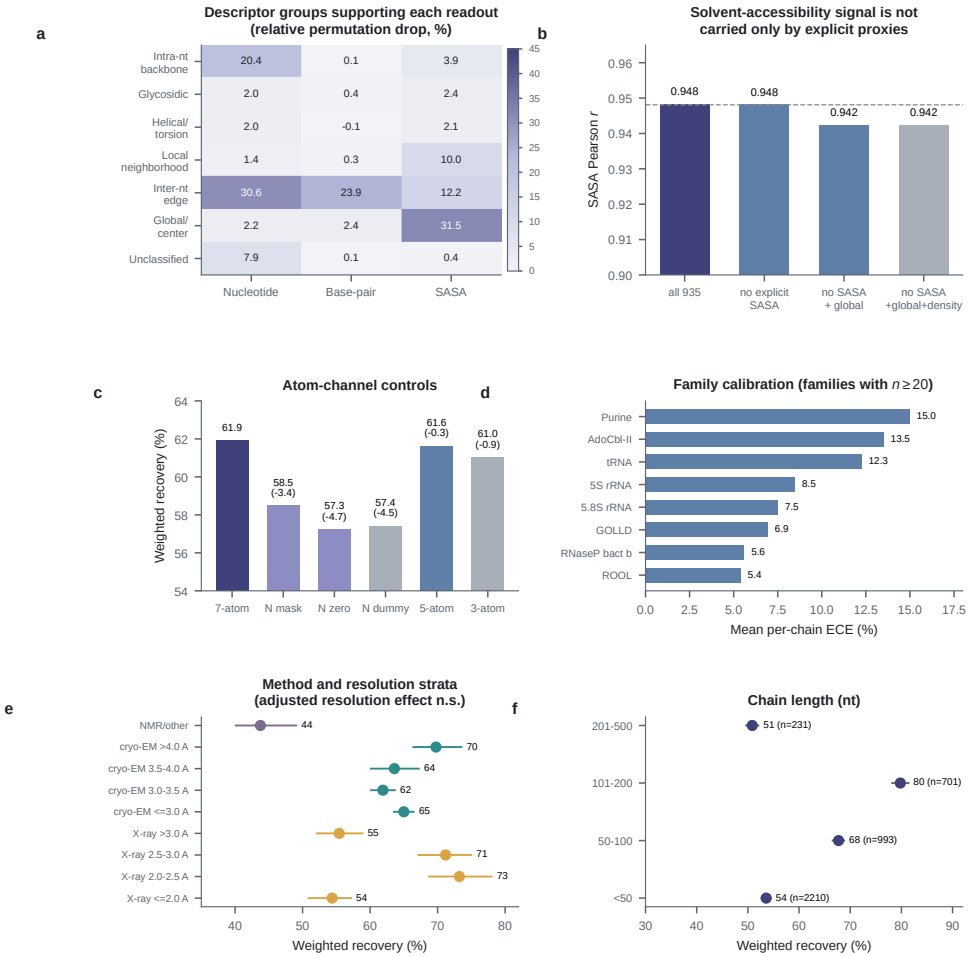



**Fig. S8 Complementary descriptor and stratum panels. a,** Grouped permutation importance. **b,** SASA ablation. **c,** Atom-channel controls. **d,** Family calibration (mean per-chain ECE). **e,** Method/resolution strata. **f,** Chain-length strata. These panels support the main information-landscape figure and are referenced where the main figure omits detail.

FASTA, not full residue-ID alignment. PDB-clustered bootstrap of the specific-versus-non-binding confidence difference gave +0.053 (95% CI −0.048 to +0.148), so the ligand-class contrast is uncertain across the 54 PDB entries.

### *Continuous stacking analysis.*

For 200 post-2025 chains (13,801 unique positions) we computed an unclipped base-plane violation for each position (centroid distance, sign-aligned normal angle and vertical offset, normalized by the label thresholds; best neighbour retained) and correlated it with FORGE confidence. Pooled Spearman was 0.008; the mean within-chain Spearman was 0.060 (95% CI 0.030–0.090); an OLS model with base and chain

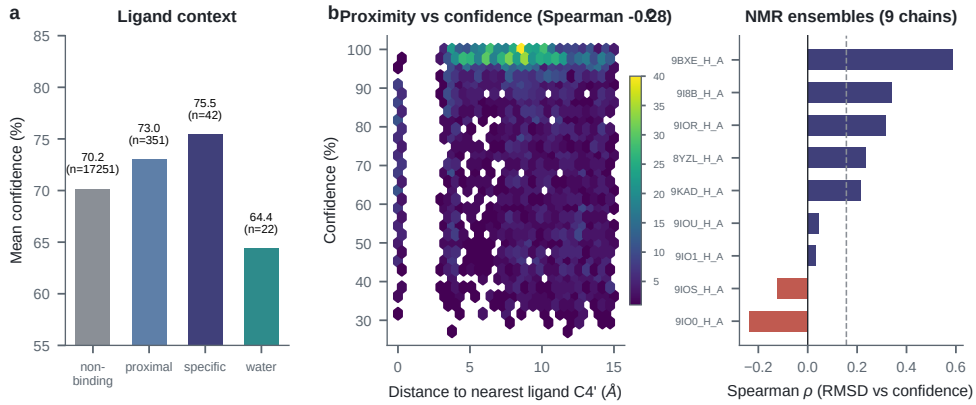



**Fig. S9 Context case studies. a,** Mean FORGE confidence for non-binding, ligand-proximal, ligand-specific and water-proximal positions in 200 post-2025 ligand-annotated chains. **b,** Hexbin of minimum ligand C4′ distance versus confidence (Spearman $\rho = -0.28$). **c,** Per-chain Spearman correlation between NMR ensemble mean per-residue P-atom displacement and FORGE confidence for nine post-2025 ensembles (mean $\rho = 0.16$; dashed line). Ligand analyses are descriptive; only modelled ligands are annotated and absence of a ligand is not evidence of non-binding.

fixed effects and chain-clustered errors gave a violation coefficient of +0.041 ($p = 2.9 \times 10^{-13}$). Greater geometric violation is therefore associated with slightly higher confidence; this is not a simple stacking-order proxy or a full physical mechanism test.

***NMR ensemble analysis.***

For post-2025 NMR entries with at least two models, per-residue P atoms present in all models were superposed by Kabsch and the mean per-residue P-atom displacement was correlated with the cached single-structure FORGE confidence; nine chains with at least five mapped A/C/G/U residues were retained. No consistent confidence-flexibility relationship was observed. This is not per-conformer inference and the residue-number mapping is not independently validated. An existing per-conformer analysis of 25 multi-model groups gave XGBoost call consistency 0.903 and a weak dispersion-inconsistency correlation ($r = 0.084$); across 11 post-2025 ensembles, accuracy was 0.352 in the high-dispersion half and 0.408 in the low-dispersion half. These are bounded descriptive results, not thermodynamic or dynamic measurements. A two-ensemble per-conformer pilot (1A60, 24 models/44-residue chain; 1BYJ, 38 models/27-residue chain; 70 positions retained after requiring all backbone atoms in every model; both pre-2025) gave accuracy 0.605, mean native-probability s.d. 0.064 and a call-switch rate of 0.255; the high-feature-variation tertile had switch rate 0.272 and probability s.d. 0.067 versus 0.175 and 0.059 in the low-feature-variation tertile. This is a pre-2025 pilot, not post-2025 generalization or a thermodynamic measurement.

The 9IWF Purine card remains a static single-structure illustration; no residue-level partner-contact/error decomposition is claimed. The protein/assembly burial and modification-ledger cases are stated boundaries, not results.

## Supplementary Table S15: validation-selected confidence threshold

| Population | Threshold | Coverage | Accuracy |
|---|---|---|---|
| Pre-2025 validation (selection) | 0.61 | 51.9% | 90.1% |
| Post-2025 all | 0.61 | 43.5% | 88.7% |
| Post-2025 repeats | 0.61 | 79.9% | 95.1% |
| Post-2025 non-repeats | 0.61 | 28.0% | 80.8% |

The threshold was selected on the first 1,000 pre-2025 validation chains (54,166 positions after cache/length filtering; not the full 3,293-chain split) as the smallest 0.01-grid value with at least 1,000 retained positions and at least 90% selective accuracy. It is applied unchanged; no post-2025 outcomes were used to choose it. PDB-clustered 95% CIs (1,000 bootstraps over 4,135 chains) were 40.8–46.3% coverage and 87.8–89.4% accuracy for all positions, 77.9–81.7% and 94.6–95.6% for repeats, and 26.3–30.0% and 79.6–82.1% for non-repeats. Coverage and accuracy differ substantially between repeats and non-repeats.

## Supplementary Table S16: common task contract

| Task | Evaluation unit | Chains | Positions | Baseline / note |
|---|---|---|---|---|
| Nucleotide identity | 4-class position | 4,135 | 291,331 | random 25%; all/repeat/joint-novel/<80% |
| Base-pair state | binary position | 2,576 | 182,338 | majority 63.1%; DSSR-eligible |
| Solvent exposure | continuous position | 4,135 | 291,331 | $r = 0$; full-atom rSASA label |
| Reactivity target | continuous position | 4,135 | 299,074 | $r = 0$; RibonanzaNet-inferred |
| Stacking | binary position | 4,135 | 290,878 | positive rate 65.7%; exploratory |
| SS class | 3-class position | 2,138 | 163,395 | loop 54.8%; class-balanced |
| Pseudoknot | binary rare class | 2,138 | 163,395 (11,828 positive) | prevalence 7.2%; $F_1 < 0.001$ at argmax |

The common contract reports the evaluation unit, denominator, baseline and novelty stratification where the underlying joins are valid. Auxiliary tasks remain exploratory and are not pooled into a single benchmark score.

## Supplementary Table S17: four-construct AF3 attribution matrix

| Construct | True P-RMSD (Å) | FORGE P-RMSD (Å) | Refined P-RMSD (Å) | True AF3-consistent? |
|---|---|---|---|---|
| 10ZT | 6.6 | – | – | yes |
| 10ZU | 14.1–15.8 | 22.7–23.6 | 19.5–28.2 | no (true control fails) |
| 11EH | 16.2–18.6 | 25.7–27.3 | 9.4–10.5 | no (true control fails) |
| 11AG | 22.1–26.9 | 18.9–22.7 | 19.1–28.2 | no (true control fails) |

Derived/refined 10ZT structures were not evaluated in this AF3 batch, so its derived rows are left blank; SS/contact provenance is reported elsewhere (Supplementary Note S1). Three of four true controls fail the AF3 test, so the assay cannot adjudicate designed-sequence foldability; it only shows that FORGE/refined readback did not yield AF3-consistent folds in this set. **11AG interface separation.** In the two-chain 11AG stress structure, 40 residues per chain lie within 8 Å of the other chain. Interface accuracy was 30.0% (chain A) and 35.0% (chain C) versus 52.5% and 50.8% outside the interface, while mean confidence was similar (0.425/0.427 versus 0.432/0.438). Interface errors are therefore not explained by lower confidence.

**Supplementary Table S18: 9IWF Purine static card**

| Quantity | Value |
|---|---|
| Positions | 69 |
| Recovery | 34.8% |
| Mean confidence | 0.476 |
| Positions with confidence $\geq 0.61$ | 1/69 |
| Accuracy, confidence 0.5–0.8 | 31.8% (22 positions) |

A GTP ligand is present in the deposited 9IWF structure: two residues (positions 1 and 65) are within 4.5 Å of GTP, with mean confidence 0.444 and accuracy 0.50 versus 0.477 and 0.343 for non-contact positions. With only two contact positions this is inconclusive, not a contact-error mechanism. The card does not include conservation or a residue-level decomposition into flexibility, missing chemistry, degeneracy and model failure. It is presented as a boundary, not as a mechanism result.